Chencheng(Charlotte) Qi

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EDUCATION

University of Florida Florida, United States
- **Master of Engineering** in Electronic Computer Engineering Sep 2023-May 2025
- **Overall GPA**: 3.7/4

Hebei University of Technology Tianjin, China
- **Master of Engineering** in Electronic Information Sep 2020-Jun 2023
- **Overall GPA**: 89.6/100
- **Publications & Patents**:
  - M. Zhao, Z. Wang, **C. Qi**, et al. Face Video Heart Rate Detection Based on FastICA and ICEEMDAN. *Chinese Journal of Biomedical Engineering*. DOI: 10. 3969 /j. ISSN. 0258-8021. 2022. 04. 014 (published)
  - **C. Qi**, M. Zhao, A Mask Occlusion Face Expression Recognition Method Based on Deep Learning. Patent 202310045293.4
  - M. Zhao, Z. Wang, C. Qi, The invention relates to a non-contact heart rate measurement method and system based on face video. Patent 202310045293.4
  - C. Qi, M. Zhao, Z. Wang. Facial expression recognition with mask occlusion. Electronic Measurement Technology. (Submitted)

Heilongjiang University Harbin, China
- **Bachelor of Engineering** in Electrical Engineering & Automation Sep 2011-Jun 2015
- **Overall GPA**: 84.6/100

RESEARCH EXPERIENCES

Machine Learning Researcher, University of Florida **Dec 2023- May 2025**

3D Brain MRI Segmentation
- Developed a preprocessing and augmentation pipeline for a 3D brain MRI dataset, incorporating artifact and noise reduction, intensity normalization, spatial resampling, and advanced data augmentation techniques to improve data quality, diversity, and model robustness.
- Enhanced an nnU-Net-based segmentation framework by integrating Swin-UNet components, attention mechanisms, and cross-channel feature weighting, achieving a 1.7-percentage-point improvement in mean Dice score over the nnU-Net baseline.

American Sign Language Character Classification
- Developed scalable Python pipelines for preprocessing, augmentation, and management of a large-scale American Sign Language image dataset containing 43,200 images (approximately 40 GB), improving data processing throughput by 30%.
- Designed and trained a ResNet50-based image classification model with integrated attention mechanisms in PyTorch, achieving 98.7% classification accuracy on the held-out test dataset.

Machine Learning Researcher, Hebei University of Technology **Sep 2020-Jun 2023**

Non-Contact Heart Rate Estimation from Facial Videos
- Developed a non-contact heart rate estimation framework that extracted physiological signals from facial videos.
- Proposed a signal-processing pipeline combining Fast Independent Component Analysis (FastICA) and Improved Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (ICEEMDAN) to isolate pulse-related signals and reduce motion and illumination interference.
- Selected physiologically relevant intrinsic mode functions (IMFs) based on expected heart-rate frequency ranges and estimated heart rate through frequency-domain spectral analysis.

Masked Facial Expression Recognition

- Constructed a real-world masked facial expression dataset by collecting images from 150 participants under controlled laboratory conditions and categorizing the samples into seven facial expression classes.
- Applied Dlib-based facial landmark detection and image reconstruction to generate realistic mask occlusions for the FER+, RaFD, and RAF-DB datasets, creating a large-scale dataset of more than 70,000 occluded facial expression images (approximately 50 GB).
- Implemented and evaluated multiple deep learning architectures, including ResNet, VGG19, Vision Transformer, and Swin Transformer, for masked facial expression recognition.
- Developed an attention-enhanced Swin Transformer model to improve the extraction of discriminative facial features under mask occlusion and designed a GAN-based image restoration framework that improved classification accuracy by 7 percentage points, achieving 90.3% accuracy.

Research Assistant, Tsinghua University **Mar 2018-Apr 2019**

Design & Implementation of Laboratory Detection Control System based on PLC

- Developed a Servo Control System based on PLC, as per the experiment table requirements.
- Confirmed controlling requirements of the device, developed PLC software, designed relevant industrial control configuration software, and debugged. Sorted out technical documents, including design instruction, electrical installation diagram, electrical components statement, and operation instruction, etc.

Undergraduate Researcher, Heilongjiang University **Oct 2014-May 2015**

Application of PVDF Piezoelectric Thin Film in Human Joint Posture Monitoring

- Proposed finger action perception algorithm based on PVDF sensor that contains signal preprocessing algorithm, finger action signal segment extraction algorithm, 4-layer BP neural network, etc.
- Implemented finger action recognition experiment through above algorithm combining PVDF sensor array.
- Concluded that the above algorithm can detect finger action efficiently and implement semantic recognition.

PROFESSIONAL EXPERIENCE

Web Developer, Morriox Group LLC **Aug 2025-Now**

- Developed and maintained full-stack web applications, including responsive user interfaces, backend services, financial data integrations, and third-party APIs. Improved system performance, scalability, and security while collaborating with cross-functional teams to support financial advisory and consulting operations.

Electrical Engineer, SINOHYDRO Bureau 3rd Co., Ltd. **Jul 2015-Mar 2018**

- Participated in designing both CEMS atmospheric smoke monitoring system and VOCS volatile organic compound detection system.
- Took responsible for debugging and maintaining the devices to keep their normal operation.
- Transferred real-time collected gas to analogue signal through analysis meter and stored these real-time data to SQL database so that the report form was generated and submitted to the environmental protection agency.

HONORS & AWARDS

- The 3rd Class Scholarship for AY 2021-2022 by Hebei University of Technology (Top 10%) 2022
- The 3rd Class Scholarship for AY 2013-2014 by Heilongjiang University (Top 5%) 2014

EXTRACURRICULAR ACTIVITY

Member of Publicity Department, College Student Union 2020-2022

- Took charge of organizing varieties of campus activities such as Campus Rainbow Running, Mid-Autumn Festival Gala, Academic Salon Lectures, etc.

Member & Minister of Publicity Dept., Student Union of Heilongjiang Univ. 2011-2013

- Organized and publicized varieties of campus activities such as the university celebration for 70th Anniversary, annual sports meetings, etc. Took the responsibilities of department managementt.

SKILLS & OTHER

- **AI/ML**: PyTorch, TensorFlow, Keras, scikit-learn, CUDA
- **Languages**: Python, Java, C, C++, SQL, MATLAB, Linux, AutoCAD
- **Standardized Tests**: TOEFL: 108 (R 30, L 30, S 22 W 26); GRE: V 160, Q 170, AW 4.5
- **Hobbies**: Painting, Photography, Hiking, etc.